

Lab 1: Product Page

Data Binding, Directives & Computed Properties

Objective

In this lab you will build a dynamic product page using Vue 3. All product data must come from the parent component only — no hardcoded text in the template. You will practice data binding, directives, and computed properties in a realistic UI scenario.

What You Will Build

Recreate the product page wireframe provided in the lecture. Your page must include:

- A main product section with image, name, description, price, discounted price, and a badge
- A tags list rendered dynamically
- A related products section rendered from an array
- Conditional display of the original price only when a discount exists

Data Shape

Define the following JSON objects in your parent component. Do not modify the structure — your template must work with this shape as-is.

Main Product

```
{
  "id": 1,
  "name": "Cozy Sneakers",
  "description": "High-quality sneakers that go with everything you wear.",
  "image": "https://via.placeholder.com/300",
  "badge": "NEW",
  "price": 120,
  "discount": 20,
  "tags": ["Fashion", "Casual", "Sport"]
}
```

Related Products

```
[
  {
    "id": 2,
    "name": "Running Shoes",
    "price": 90,
    "discount": 10,
    "image": "https://via.placeholder.com/150"
  },
  {
    "id": 3,
    "name": "Casual Boots",
    "price": 150,
    "discount": 0,
    "image": "https://via.placeholder.com/150"
  },
  {
    "id": 4,
    "name": "Flip Flops",
    "price": 30,
    "discount": 50,
    "image": "https://via.placeholder.com/150"
  }
]
```

Requirements

You must use the following Vue features. Each one must appear in your template at least once:

Directive / Feature	Use it for	Notes
<code>{{ }}</code>	Product name, description, price, badge	
<code>:src</code>	Product image binding	
<code>:class / :style</code>	Badge color (conditional)	
<code>v-for</code>	Tags list and related products	
<code>v-if</code>	Show original price only if discount > 0	
<code>computed</code>	Calculate discounted price	

Important Rules

- All data lives in the parent component only — no data defined inside a child component for this lab
- No hardcoded text in the template — every visible value must come from your data
- The discounted price must be calculated using a computed property, not manually in the template
- If a related product has a discount of 0, do not show a strikethrough price for it
- Use DaisyUI or Bootstrap for styling — your page should look presentable, not bare HTML

Acceptance Criteria

Your lab is complete when:

1. Changing the price or discount value in the script instantly updates the displayed prices
2. Related products render from the array — adding or removing an item from the array reflects in the UI
3. The badge only shows when a badge value exists
4. Tags render as individual elements, not as a plain comma-separated string
5. The original price is visible with a strikethrough only when discount is greater than zero

Bonus Challenge

If you finish early: Add an **isAvailable** boolean to the product JSON. Show an "Out of Stock" badge and disable the buy button when it is false.

Props and component communication will be covered in Day 2.